pfi

Python for Fixed Income

COURSE 7 · NLP, RAG, AND LLMS

Covenants 2: Baskets, Carve-Outs, and Two Vintages

WEEK 3 · DAY 3

Created by **Jeff Lenamon**

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Today: two indentures and a PDF

Three documents

Boyd 2021	4.750% notes due 2031, 8-K Ex. 4.1
Levi 2006	8-7/8% notes due 2016, 8-K Ex. 4.1
Boyd PDF	printed by the course with Edge
Saved	responses recorded 2026-10-08

Today

1. How a covenant is built
2. Grower baskets: regex and model
3. Permitted Liens; the builder basket
4. The second vintage, same checks
5. One comparable table; a word diff
6. PDF: text layer, then page image
7. Which route when; Excel; AI check

A regular expression and a model read the same baskets, and each one checks the other.

The rules for real documents

Every prompt asks what a document says: terms as JSON, a quote beside every number

Allowed names

the indenture states

extracted value, with its quote

passed the source check

Never called

loose, tight, weak, strong

covenant-lite or protective

a credit view or recommendation

A comparison states numbers and text, never quality: the table has no column that grades.

How a covenant is built

Prohibition	the Company shall not ... Incur any Indebtedness
Ratio debt	unless its Consolidated Fixed Charge Coverage Ratio would have been at least 2.0 to 1.0
Permitted debt	clause (b): 17 clauses, each a basket: (iii) credit facility, (ix) capex, (xv) general
Carve-outs	clause (d): accrued interest, accretion, PIK interest are not an Incurrence

Source: Boyd Gaming Corporation, indenture for the 4.750% Senior Notes due 2031, Exhibit 4.1 to the 8-K filed 2021-06-08, accession 0001193125-21-185495, Section 4.12.

How a grower basket is read

The problem	One sentence holds two numbers and a defined term
The pattern	A regular expression finds each amount, then the greater-of wording
No guessing	A sentence the pattern does not match is not returned
What you control	The text and, if you want, the pattern
What comes out	A row per amount: kind, share, defined term, section

Grower baskets, read by a regular expression

57

dollar amounts in the Boyd text, basket_amounts

6

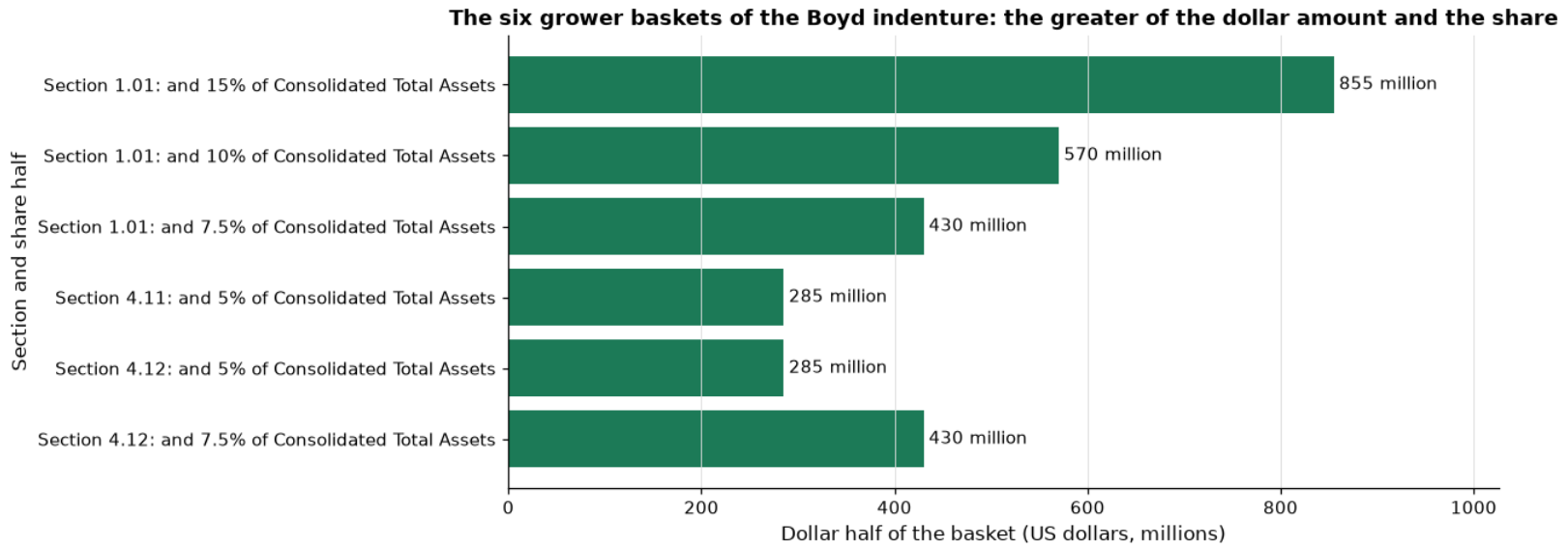
growers: the greater of $X and Y% of a defined term

```
Section 1.01: the greater of $855,000,000 and 15% of CTA (Permitted Investment)
Section 1.01: the greater of $570,000,000 and 10% of CTA (Permitted Investment)
Section 1.01: the greater of $430,000,000 and 7.5% of CTA (Permitted Liens)
Section 4.11: the greater of $285,000,000 and 5% of CTA
Section 4.12: the greater of $285,000,000 and 5% of CTA
Section 4.12: the greater of $430,000,000 and 7.5% of CTA
```

CTA is Consolidated Total Assets. The credit facility's "$3.5 billion, and 4.5 times EBITDA" reads as an amount.

Source: Boyd Gaming Corporation, indenture for the 4.750% Senior Notes due 2031, Exhibit 4.1 to the 8-K filed 2021-06-08, accession 0001193125-21-185495.

Grower baskets, drawn



6

growers read by basket_amounts in the Boyd text

4.11 and 4.12

write 285,000,000 dollars and 5%

The figure shows how the baskets are written. Which half is larger depends on total assets, which no bar shows.

The model's baskets beside the regex

6 of 7

passed, Section 4.12, saved response

4 of 6

Boyd growers with a basket of the same numbers

3

of those citing the grower's own section

```
Credit Facilities: percent 4.5 4.12 yes yes no no
```

The 4.5 is "4.5 times Consolidated EBITDA": a multiple in the percent field. Day 12's third change had taken it out; today's growers sentence put it back.

A new sentence can undo what an earlier one fixed. The check is what shows it.

Permitted Liens and the builder basket

Section 4.14: Permitted Liens

38 clauses, (a) to (ll), in Section 1.01

(jj): the greater of $430,000,000

and 7.5% of Consolidated Total Assets

Saved response: cited 1.01, passed

Day 12's: cited 4.14, quote failed

Section 4.15: restricted payments

Builder: 50% of Consolidated Net Income

from April 1, 1997, plus (B) to (F)

(b)(xii): $400,000,000 general basket

(b)(xiii): leverage not over 4.00 to 1.00

Saved response: 50% and $50,000,000

The $400,000,000 is in no saved response, day 12's or today's; basket_amounts reads it in the text.

How a second indenture goes through

The problem	Is the schema built around one document?
Nothing built in	Prompt, schema, and checks are handed a text and its sections
What changes	The names: Levi numbers its covenants differently
What you control	The text, the section numbers, the kinds
What comes out	A second package and a second checked table

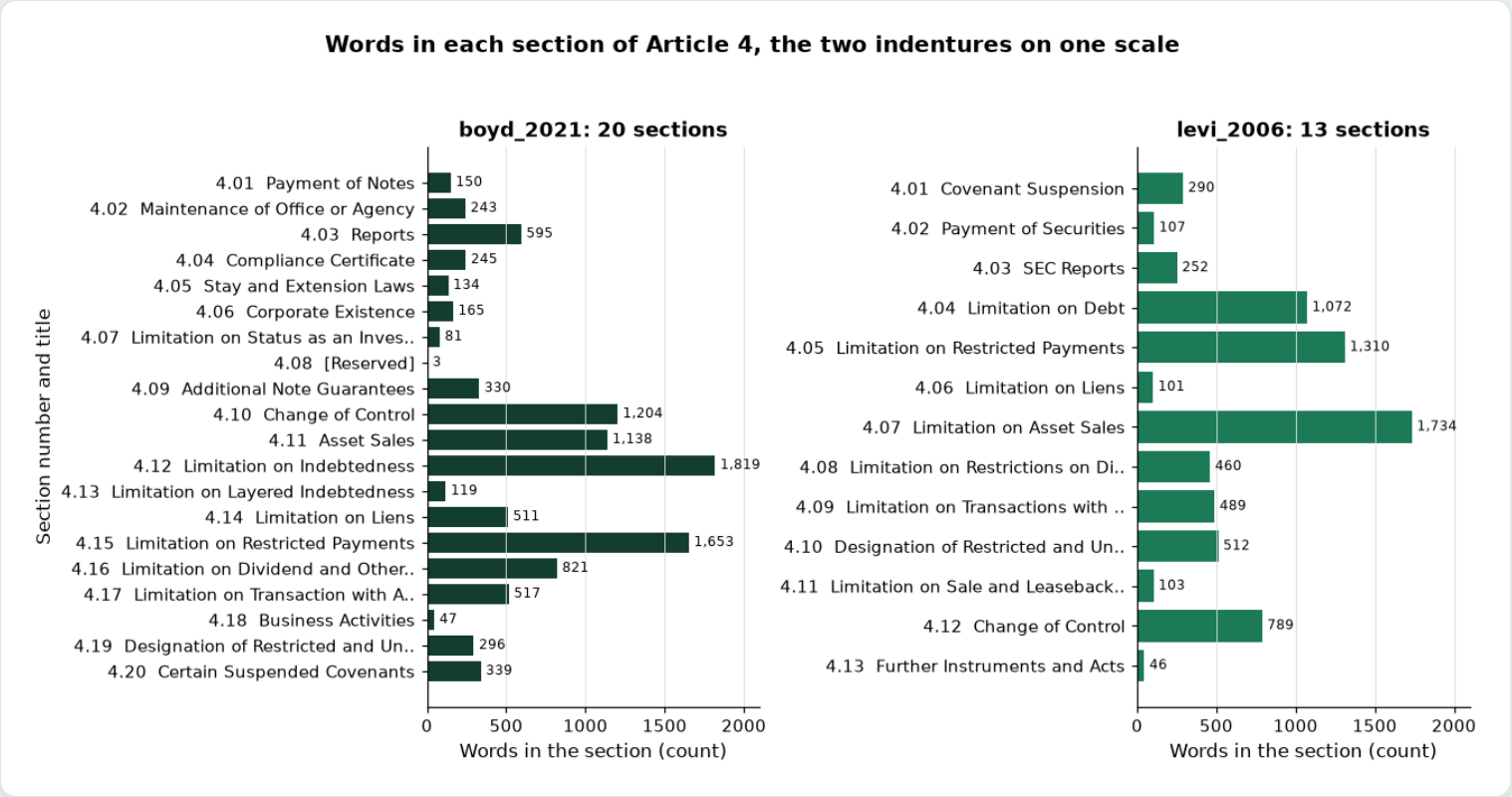
The second vintage

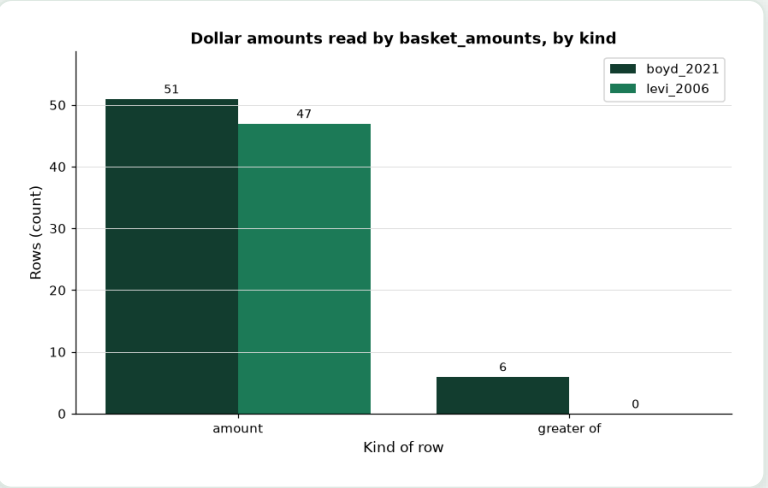
	Boyd 2021	Levi 2006
Article 4	20 sections; 4.12 Limitation on Indebtedness	13 sections; 4.04 Limitation on Debt
Ratio	Fixed Charge, at least 2.0 to 1.0	Fixed Charges, greater than 2.00 to 1.00
Permitted debt	clause (b), 17 clauses	"Permitted Debt" in 4.04, 12 clauses
Liens list	38 clauses, (jj) a grower	20 clauses, (t) 5.0% only
basket_amounts	57 amounts, 6 growers	47 amounts, no grower

Levi's credit facility basket: "the greater of: (1) $1.6 billion, which amount shall be permanently reduced" and (2) 50% of inventory plus 85% of receivables.

Source: Levi Strauss & Co., indenture for the 8-7/8% Senior Notes due 2016, Exhibit 4.1 to the 8-K filed 2006-03-17, accession 0000950134-06-005445, Section 4.04.

The two Article 4s, drawn





Levi writes its greater-of basket in another shape than the pattern reads, so its $1.6 billion counts as a plain amount.

Both panels share one scale: the covenants a desk records are the long sections in both.

Levi through the same schema and checks

3 of 4

covenants parse, saved responses

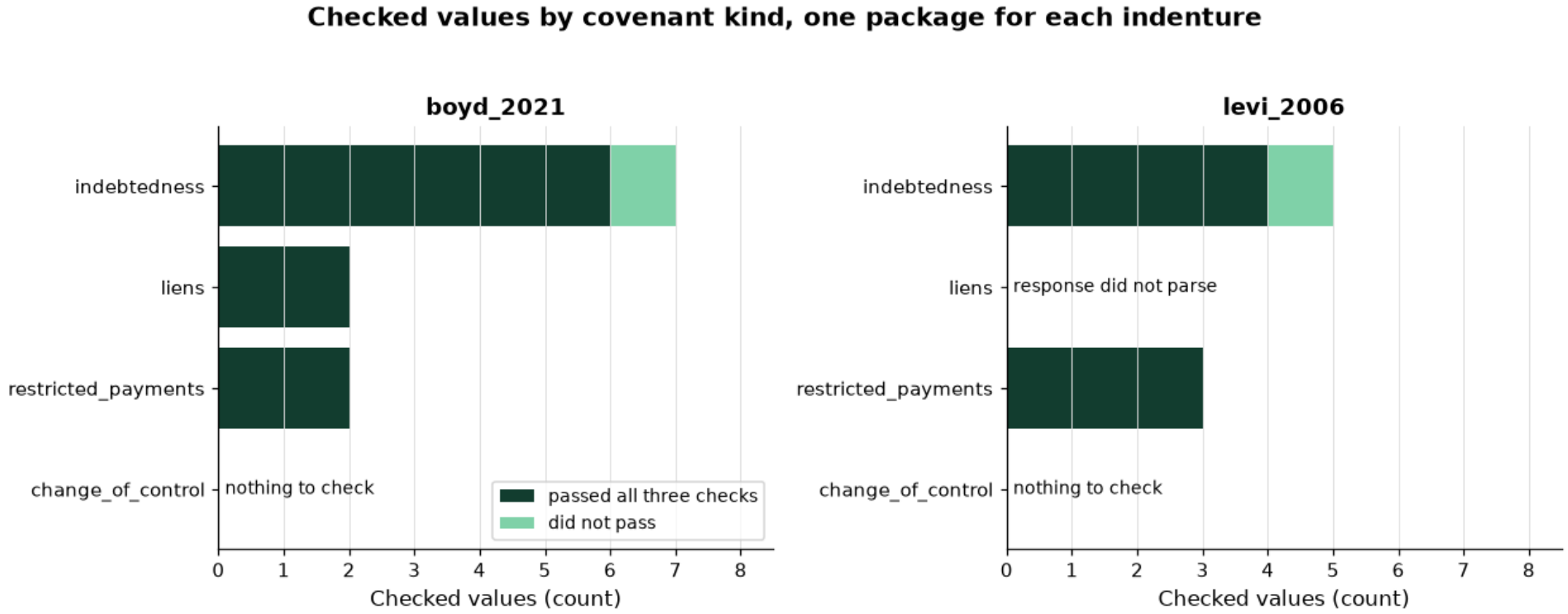
7 of 8

values passed all three checks

4.06	the response does not fit Covenant: quote: Field required
$1.6 billion	one basket came back as three rows: $1.6 billion, 50%, 85%
Rejected	"Credit 41 Facilities": the page number sits inside the quoted sentence

Same checks for both indentures: neither document gets a special rule.

Checks by covenant kind, both indentures



7 of 8

Levi values passed

Two documents read through one schema, side by side. A count of checked values is not a ranking.

How the comparable table is made

The problem	Two documents, one question: what does each say?
The rows	One row for each covenant kind and field
The columns	A value and a section for each indenture
No judgement	A gap is an empty cell; no difference is computed
What comes out	A table of what each document says

One comparable table

compare_packages: 25 rows; kind | field | Boyd value, section | Levi value, section

```
indebtedness title | Limitation on Indebtedness 4.12 | Limitation on Debt 4.04
indebtedness ratio_test | Consolidated Fixed Charge Coverage Ratio, at 4.12 | Consolidated Fixed
indebtedness basket 3: other Indebtedness | the greater of $430,000,000 and 7.5% of Comp 4.12 |
indebtedness basket 4: General basket | | $100,000,000 4.04
liens title | Limitation on Liens 4.14 |
restricted_payments title | Limitation on Restricted Payments 4.15 | Limitation on Restricted Pa
change_of_control title | Change of Control 4.10 | Change of Control 4.12
```

```
[-boyd_2021-]{+levi_2006+},indebtedness,ratio_test.threshold,2.0,x,[-4.12-]{+4.04+},True,True,nv
```

No column of judgement: Boyd's general basket is a grower; Levi's is $100,000,000. The table states both.

How a PDF page is read

The problem	A PDF page can hold text, pixels, or both
Text layer	Characters with positions; pypdf restores the order
Pixels	A scan: draw the page, send it to a model, check the rows
What you control	The page, and for an image the scale and the prompt
What comes out	The page text, or the model's rows from the picture

PDF pages: the text layer

The Boyd indenture printed to PDF by the course from the SEC copy with Edge: the SEC's text, Edge's layout.

132

pages in the PDF

[51, 109]

pages holding the redemption table

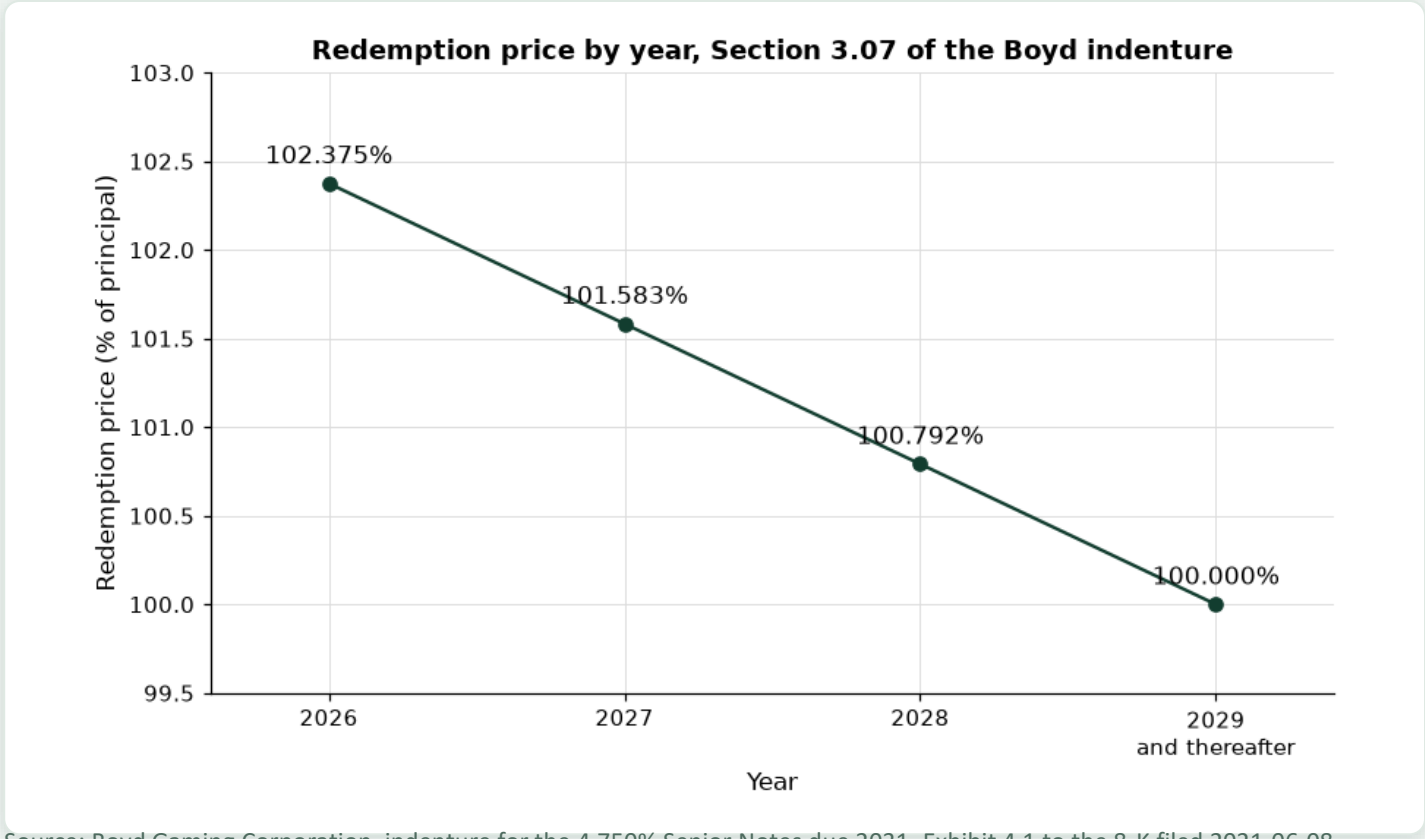
47

last line: the printed page number

```
2026 HTML 102.375% PDF text layer 102.375% equal: yes
2027 HTML 101.583% PDF text layer 101.583% equal: yes
2028 HTML 100.792% PDF text layer 100.792% equal: yes
2029 and thereafter HTML 100.000% PDF text layer 100.000% equal: yes
```

Source: Boyd Gaming Corporation, indenture for the 4.750% Senior Notes due 2031, Exhibit 4.1 to the 8-K filed 2021-06-08, accession 0001193125-21-185495, Section 3.07.

The redemption table as numbers

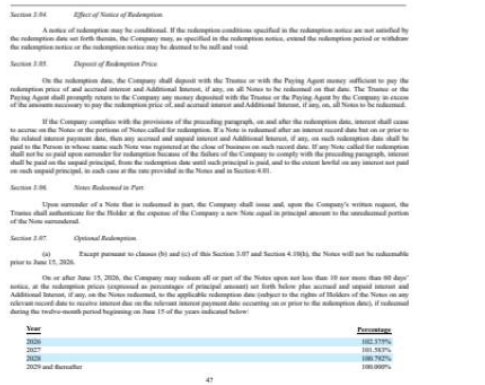


Year	Price (%)	Step
2026	102.375	-
2027	101.583	-0.792
2028	100.792	-0.791
2029 and thereafter	100.000	-0.792

Source: Boyd Gaming Corporation, indenture for the 4.750% Senior Notes due 2031, Exhibit 4.1 to the 8-K filed 2021-06-08, accession 0001193125-21-185495, Section 3.07.

The page as an image

Page 51 of the PDF, rendered by page_png



Section 3.04 *Effect of Notice of Redemption.*

A notice of redemption may be conditional. If the redemption conditions specified in the redemption notice are not satisfied by the redemption date set forth therein, the Company may, as specified in the redemption notice, extend the redemption period or withdraw the redemption notice or the redemption notice may be deemed to be null and void.

Section 3.05 *Deposit of Redemption Price.*

On the redemption date, the Company shall deposit with the Trustee or with the Paying Agent money sufficient to pay the redemption price of and accrued interest and Additional Interest, if any, on all Notes to be redeemed on that date. The Trustee or the Paying Agent shall promptly return to the Company any money deposited with the Trustee or the Paying Agent by the Company in excess of the amounts necessary to pay the redemption price of, and accrued interest and Additional Interest, if any, on, all Notes to be redeemed.

If the Company complies with the provisions of the preceding paragraph, on and after the redemption date, interest shall cease to accrue on the Notes or the portions of Notes called for redemption. If a Note is redeemed after an interest record date but on or prior to the related interest payment date, then any accrued and unpaid interest and Additional Interest, if any, on such redemption date shall be paid to the Person in whose name such Note was registered at the close of business on such record date. If any Note called for redemption shall not be so paid upon surrender for redemption because of the failure of the Company to comply with the preceding paragraph, interest shall be paid on the unpaid principal, from the redemption date until such principal is paid, and to the extent lawful on any interest not paid on such unpaid principal, in each case at the rate provided in the Notes and in Section 4.01.

Section 3.06 *Notes Redeemed in Part.*

Upon surrender of a Note that is redeemed in part, the Company shall issue and, upon the Company's written request, the Trustee shall authenticate for the Holder at the expense of the Company a new Note equal in principal amount to the unredeemed portion of the Note surrendered.

Section 3.07 *Optional Redemption.*

(a) Except pursuant to clauses (b) and (c) of this Section 3.07 and Section 4.10(k), the Notes will not be redeemable prior to June 15, 2026.

On or after June 15, 2026, the Company may redeem all or part of the Notes upon not less than 10 nor more than 60 days' notice, at the redemption prices (expressed as percentages of principal amount) set forth below plus accrued and unpaid interest and Additional Interest, if any, on the Notes redeemed, to the applicable redemption date (subject to the rights of Holders of the Notes on any relevant record date to receive interest due on the relevant interest payment date occurring on or prior to the redemption date), if redeemed during the twelve-month period beginning on June 15 of the years indicated below:

Year	Percentage
2026	102.375%
2027	101.583%
2028	100.792%
2029 and thereafter	100.000%

47

PNG of 289,865 bytes, no text layer; 2,119 prompt tokens; free

```
2026 HTML 102.375% image 2026 102.375% equal: yes
2027 HTML 101.583% image 2027 101.583% equal: yes
2028 HTML 100.792% image 2028 100.792% equal: yes
2029 and thereafter HTML 100.000% image 2029 and thereafter 100.000% equal: yes
```

Model nvidia/nemotron-3-nano-omni-30b-a3b-reasoning:free; same table as the HTML: yes

Replay needs the exact request: the cell uses the image the course sent, and says whether this render equals it.

Source: Boyd Gaming Corporation, indenture for the 4.750% Senior Notes due 2031, Exhibit 4.1 to the 8-K filed 2021-06-08, accession 0001193125-21-185495, Section 3.07.

Which route when

Route	Function	Rows	Requests
Filed HTML, as text	html_to_text, section_text	4, exact	0
PDF text layer	pdf_text	4, equal; page number inside	0
Page as an image	page_png, image_message, ask	4, equal: yes	1

The text layer first. A page image only when there is no text layer, or it is out of order.

Text can be checked character for character; what a model returns from pixels can only be checked against text.

How a message and its tokens work

The problem	What goes in a request, and what does it cost?
Content parts	A string, or a list of parts, each with a kind
Tokens	Every part becomes tokens, counted going in and coming back
What you control	The parts and, for an image, its detail
What comes out	A response whose usage reports both counts

The kinds of content part

text (course sends it: yes, in every request)

Plain text: the instruction, or a document.

image_url (course sends it: yes, once (section 13.7))

An image, as a web address or as a data: URL that holds the image's bytes.

input_audio (course sends it: no)

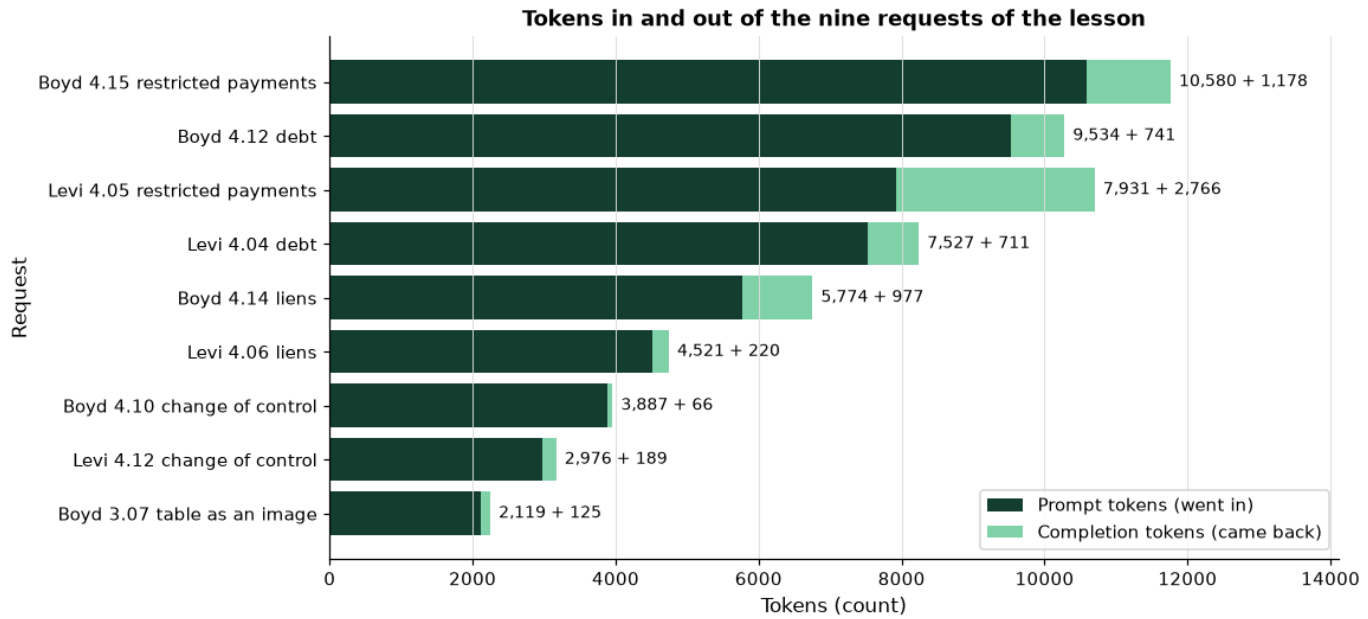
Audio bytes and their format.

file (course sends it: no)

A file sent with the request.

An image_url part also takes detail; the course leaves it unset, so the provider's default applies.

Tokens in and out of nine requests



10,580

prompt tokens: Boyd 4.15 restricted payments

2,119

prompt tokens: the page as an image

The prompt side is the document; the completion side is the answer.

A page image costs about as many tokens as a short section of text: it suits one table, not a whole indenture.

Excel: baskets and the checked table

Excel side by side (tested in Excel)

`=TEXTBEFORE(TEXTAFTER(A2,"greater of "),"%")  clause (jj)`	$430,000,000 and 7.5
`=TEXTBEFORE(TEXTAFTER(A2,"greater of "),"  and ")  clause (jj)`	$430,000,000
`=TEXTBEFORE(TEXTAFTER(A2,"greater of "),"%")  Levi (b), "greater of:"`	#N/A
`=LEN(C2)<>LEN(SUBSTITUTE(C2,B2,""))  quotes that fit a cell`	19 of 19 equal to quote_found
`=LEN(C2)  C2 = Section 1.01`	32,767 of 107,255: SUBSTITUTE FALSE

Check a quote from Section 1.01 against its definition. pandas read 4.10 back as 4.1; dtype=str keeps '4.10'.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

Units, conventions, and what the function must never do: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Assert what the docstring promises. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: a basket regex reads "$855,000,000" as 855 and "$1.6 billion" as 1, so a right amount looks wrong.

Code that runs is not the same as code that is right.

What you can do now

- Read a covenant: prohibition, ratio debt, baskets, carve-outs
- Cross-check the model's baskets with basket_amounts
- Run a second vintage through the same schema and checks
- compare_packages: values and sections, no judgement
- PDF: text layer first, a page image checked against text

Week 3, Day 4 RAG over an indenture